

Radical S&P 500 Astrology Event Study

Dogfooding hermetic-alpha-library v0.1.1 on ^GSPC, 1927-2026

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Daftar Isi

1	Radical S&P 500 Astrology Event Study	2
1.1	Executive summary	2
1.2	Research question	2
1.3	Data dan metodologi	2
1.4	Baseline S&P 500	4
1.5	Results: feature-level findings	4
1.6	Outer Big Cycles	4
1.7	No-Moon Full Scan	6
1.8	Deep dives on the most important patterns	7
1.9	Theme-level read	12
1.10	Interpretasi radikal tapi disiplin	14
1.11	Anti-overfitting notes	15
1.12	Working theory untuk SPX	15
1.13	Next research steps	15
1.14	Appendix A. Output artifacts	15
1.15	Disclaimer	16

1 Radical S&P 500 Astrology Event Study

Repository: wauputr4/hermetic-alpha-library

Library: hermetic-alpha v0.1.1 + pyswisseph backend 20230604

Asset: ^GSPC / S&P 500 index proxy via Yahoo Finance

Actual data: 1927-12-30 → 2026-06-04 (24,723 daily candles)

Train/test split: train through 1987-12-31 (15,045 candles), test after (9,678 candles)

Status: eksplorasi statistik / dogfood library, bukan financial advice.

1.1 Executive summary

Penelitian ini menguji apakah window aspek astrologi besar punya korelasi historis dengan forward return S&P 500. Metodenya sengaja dibuat lebih ketat dari sekadar menghitung setiap hari aspek aktif: semua active aspect day dikelompokkan menjadi window, lalu hanya **nearest exact date** yang dipakai sebagai event. Untuk kalender saham, event yang jatuh weekend/libur dipetakan ke trading day berikutnya maksimum 3 hari.

Temuan utamanya:

1. **S&P 500 punya drift positif kuat**, jadi pola harus dibandingkan melawan baseline asset sendiri. Baseline 180d rata-rata adalah **5.731%** dengan bullish rate **68.728%**.
2. **Saturn-Neptune square** muncul sebagai kandidat konstruktif paling menarik pada horizon 180d: 12-13 event, all-events avg sekitar **12.6-13.6%**, bullish **100%**, train/test edge sama-sama positif. Namun sample kecil, jadi ini hipotesis siklus besar, bukan sinyal trading.
3. **Saturn-Neptune trine, Jupiter-Uranus sextile**, dan **Saturn-Uranus square** muncul sebagai pressure / underperformance candidates, terutama 90-180d. Beberapa terlihat regime-sensitive; post-GFC tidak selalu seburuk era sebelumnya.
4. Pola BTC tidak transfer satu-ke-satu ke SPX. Theme venus_opposition_risk yang menarik di BTC justru mild underperformance di SPX pada 14d. Sebaliknya, theme cooldown mercury_venus cukup konsisten sebagai pause window.
5. Dalam scan no-moon full, Mars opposition ke Sun/Mercury/Venus terlihat konstruktif di 180d — berlawanan dengan tafsir simbolik simplistik bahwa Mars opposition otomatis bearish.

1.2 Research question

Pertanyaan risetnya: jika hermetic-alpha-library dipakai untuk event study finansial, apakah aspek planet besar bisa membentuk bucket return yang **konsisten train/test**, bukan cuma cherry-picked? Fokusnya bukan membuktikan astrologi sebagai kausalitas market, melainkan mengubah klaim yang biasanya naratif menjadi objek data: event date, forward return, baseline, sample size, dan validation split.

1.3 Data dan metodologi

1.3.1 Data market

- Symbol Yahoo: ^GSPC
- Requested start: 1900-01-01
- Actual start: 1927-12-30
- Actual end: 2026-06-04

- Candle count: 24,723
- Caveat: data Yahoo untuk indeks panjang historis berguna untuk eksplorasi, tapi bukan audit-grade dan bukan instrumen investable langsung seperti SPY/ES futures.

1.3.2 Event construction

1. Generate posisi planet harian untuk rentang candle.
2. Scan aspek: conjunction, opposition, trine, square, sextile.
3. Kelompokkan active aspect days berdasarkan feature `body_a_body_b_aspect`.
4. Collapse consecutive days menjadi satu window.
5. Pilih exact event date = baris dengan orb terkecil di window itu.
6. Untuk ^GSPC, jika exact date bukan trading day, map ke same/next available trading day dalam 3 hari kalender.
7. Hitung forward return 3d, 7d, 14d, 30d, 60d, 90d, 180d.

1.3.3 Universes

1.3.3.1 Outer Big Cycles

- Internal name: `spx_outer_big_cycles`
- Bodies: jupiter, saturn, uranus, neptune, pluto
- Aspects: conjunction, opposition, trine, square, sextile
- Orb: 3.0°
- Horizons: 3d, 7d, 14d, 30d, 60d, 90d, 180d
- Raw aspect days: 56,010
- Exact windows mapped: 661
- Features scored: 24
- Tags/themes scored: 29/3

1.3.3.2 No-Moon Full Scan

- Internal name: `spx_no_moon_full`
- Bodies: sun, mercury, venus, mars, jupiter, saturn, uranus, neptune, pluto
- Aspects: conjunction, opposition, trine, square, sextile
- Orb: 2.5°
- Horizons: 3d, 7d, 14d, 30d, 60d, 90d, 180d
- Raw aspect days: 145,527
- Exact windows mapped: 18,342
- Features scored: 145
- Tags/themes scored: 92/12

1.4 Baseline S&P 500

Ini baseline forward return semua candle valid. Semua pola harus dinilai relatif terhadap angka ini, bukan relatif terhadap nol.

Horizon	N	Avg	Median	Bullish	Min	Max
3d	24,720	0.094%	0.183%	55.174%	-26.338%	22.655%
7d	24,716	0.219%	0.395%	56.987%	-29.161%	29.642%
14d	24,709	0.440%	0.737%	59.225%	-37.531%	55.439%
30d	24,693	0.951%	1.390%	61.013%	-41.446%	78.750%
60d	24,663	1.902%	2.416%	63.824%	-50.113%	93.435%
90d	24,633	2.825%	3.545%	65.392%	-49.024%	117.179%
180d	24,543	5.731%	6.760%	68.728%	-65.135%	84.891%

1.5 Results: feature-level findings

1.6 Outer Big Cycles

1.6.1 Top constructive candidates

#	Feature	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Exact-date span
1	neptune_saturn_square	13	180d	12.563%	11.707%	100.000%	+8.752 pp	+4.594 pp	1944-07-02 → 2016-09-10
2	jupiter_saturn_sextile	22	180d	8.947%	11.153%	90.909%	+0.645 pp	+6.930 pp	1937-04-03 → 2024-02-06
3	jupiter_saturn_square	27	60d	4.658%	5.022%	77.778%	+2.592 pp	+2.962 pp	1935-11-26 → 2025-06-15
4	jupiter_neptune_square	26	90d	5.235%	5.943%	84.615%	+2.124 pp	+2.867 pp	1929-06-07 → 2025-06-19
5	pluto_saturn_sextile	13	30d	2.922%	1.332%	84.615%	+0.885 pp	+2.902 pp	1942-06-09 → 2026-03-28
6	jupiter_pluto_square	25	30d	2.218%	1.390%	80.000%	+0.582 pp	+2.293 pp	1928-03-31 → 2023-05-17
7	neptune_pluto_sextile	45	90d	4.620%	4.411%	75.000%	+2.379 pp	+0.237 pp	1945-02-19 → 2026-06-03
8	jupiter_pluto_trine	26	14d	1.293%	1.800%	73.077%	+1.069 pp	+0.556 pp	1935-03-13 → 2024-06-02

#	Feature	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Exact-date span
9	jupiter_saturn_trine	30	30d	1.584%	1.687%	66.667%	+0.643 pp	+0.619 pp	1928-04-16 → 2025-11-17
10	pluto_saturn_square	11	7d	0.637%	0.120%	54.545%	+0.665 pp	+0.120 pp	1939-07-28 → 2010-08-21

1.6.2 Top pressure / underperformance candidates

#	Feature	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Exact-date span
1	neptune_saturn_trine	16	180d	-3.238%	-5.486%	43.750%	-10.244 pp	-6.162 pp	1929-03-06 → 2013-07-19
2	jupiter_uranus_sextile	22	90d	-3.206%	-2.156%	36.364%	-5.580 pp	-6.998 pp	1929-08-07 → 2022-02-17
3	saturn_uranus_square	16	180d	-2.625%	2.254%	56.250%	-11.837 pp	-2.552 pp	1930-02-22 → 2022-10-04
4	jupiter_pluto_opposition	12	90d	-2.851%	-1.227%	50.000%	-5.171 pp	-6.384 pp	1937-05-29 → 2014-01-31
5	jupiter_uranus_square	24	180d	-1.260%	1.386%	54.167%	-8.443 pp	-4.085 pp	1930-09-05 → 2021-01-17
6	pluto_saturn_trine	13	180d	0.392%	3.794%	61.538%	-8.987 pp	-1.081 pp	1937-03-26 → 2008-04-26
7	jupiter_uranus_trine	27	180d	0.712%	3.020%	62.963%	-7.017 pp	-2.112 pp	1931-10-10 → 2019-12-15
8	jupiter_saturn_opposition	17	90d	-2.055%	-2.724%	47.059%	-6.797 pp	-2.142 pp	1930-07-27 → 2012-01-09
9	jupiter_uranus_opposition	15	180d	1.720%	4.200%	53.333%	-4.182 pp	-3.816 pp	1934-01-30 → 2017-09-28
10	jupiter_neptune_sextile	24	60d	-0.771%	-1.897%	37.500%	-2.530 pp	-2.872 pp	1930-07-03 → 2024-05-23

1.7 No-Moon Full Scan

1.7.1 Top constructive candidates

#	Feature	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Exact-date span
1	neptune_saturn_square	12	180d	13.591%	12.647%	100.000%	+8.752 pp	+6.613 pp	1944-07-02 → 2016-09-10
2	jupiter_uranus_conjunction	13	90d	7.507%	8.256%	76.923%	+3.194 pp	+7.061 pp	1928-01-25 → 2024-04-20
3	jupiter_saturn_sextile	22	180d	10.881%	11.153%	95.455%	+4.223 pp	+6.265 pp	1937-12-12 → 2024-02-06
4	mars_sun_opposition	46	180d	9.319%	12.197%	80.435%	+3.311 pp	+4.021 pp	1928-12-21 → 2025-01-16
5	jupiter_saturn_square	26	60d	4.833%	5.275%	76.923%	+2.592 pp	+3.393 pp	1935-11-26 → 2025-06-15
6	mars_mercury_opposition	49	180d	8.667%	10.207%	77.551%	+2.809 pp	+3.137 pp	1928-12-20 → 2025-01-23
7	mars_neptune_conjunction	53	60d	3.979%	3.407%	75.000%	+1.009 pp	+3.786 pp	1929-07-03 → 2026-04-13
8	mars_venus_opposition	45	180d	8.129%	10.333%	80.000%	+1.926 pp	+3.109 pp	1928-11-23 → 2024-12-12
9	mars_pluto_conjunction	55	90d	4.628%	6.233%	70.370%	+0.476 pp	+4.059 pp	1929-04-16 → 2026-01-27
10	jupiter_neptune_square	27	90d	5.188%	5.299%	85.185%	+2.124 pp	+2.710 pp	1929-06-07 → 2025-06-19

1.7.2 Top pressure / underperformance candidates

#	Feature	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Exact-date span
1	neptune_saturn_trine	16	180d	-3.238%	-5.486%	43.750%	-10.244 pp	-6.162 pp	1929-03-06 → 2013-07-19

#	Feature	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Exact-date span
2	saturn_uranus_square	17	180d	-2.400%	1.330%	58.824%	-11.837 pp	-2.835 pp	1930-02-22 → 2022-10-04
3	jupiter_uranus_sextile	22	90d	-3.206%	-2.156%	36.364%	-5.580 pp	-6.998 pp	1929-08-07 → 2022-02-17
4	jupiter_uranus_square	23	180d	-0.610%	2.620%	56.522%	-7.543 pp	-4.085 pp	1930-09-05 → 2021-01-17
5	jupiter_pluto_opposition	13	90d	-2.105%	0.258%	53.846%	-5.171 pp	-4.649 pp	1937-05-29 → 2014-04-20
6	jupiter_uranus_opposition	14	180d	1.360%	1.609%	50.000%	-4.182 pp	-4.622 pp	1934-01-30 → 2017-09-28
7	jupiter_saturn_opposition	18	90d	-2.678%	-2.767%	44.444%	-7.642 pp	-2.142 pp	1930-07-27 → 2012-01-09
8	pluto_saturn_trine	13	180d	0.392%	3.794%	61.538%	-8.987 pp	-1.081 pp	1937-03-26 → 2008-04-26
9	neptune_saturn_sextile	11	90d	-0.850%	-0.582%	45.455%	-2.514 pp	-5.069 pp	1946-11-02 → 2019-11-09
10	jupiter_uranus_trine	28	180d	1.398%	4.063%	64.286%	-5.769 pp	-2.112 pp	1931-10-10 → 2019-12-15

1.8 Deep dives on the most important patterns

1.8.1 Constructive candidates

1.8.1.1 neptune_saturn_square

- **Events:** 13 total; 7 train, 6 test
- **Exact-date span:** 1944-07-02 → 2016-09-10
- **Mapped weekend/holiday events:** 4
- **Best horizon:** 180d
- **All events:** avg 12.563%, median 11.707%, bullish 100.000%; baseline avg 5.731%
- **Train edge:** +8.752 pp; **test edge:** +4.594 pp

Regime read:

- early_1927_1971: events 5; 180d avg 14.095%, median 13.587%, bullish 100.000%

- post_bretton_woods_1971_2008: events 5; 180d avg 15.340%, median 15.760%, bullish 100.000%
- post_gfc_2009_now: events 3; 180d avg 5.384%, median 4.212%, bullish 100.000%

1.8.1.2 jupiter_saturn_sextile

- **Events:** 22 total; 13 train, 9 test
- **Exact-date span:** 1937-04-03 → 2024-02-06
- **Mapped weekend/holiday events:** 12
- **Best horizon:** 180d
- **All events:** avg 8.947%, median 11.153%, bullish 90.909%; baseline avg 5.731%
- **Train edge:** +0.645 pp; **test edge:** +6.930 pp

Regime read:

- early_1927_1971: events 9; 180d avg 7.269%, median 12.978%, bullish 88.889%
- post_bretton_woods_1971_2008: events 9; 180d avg 8.389%, median 10.040%, bullish 88.889%
- post_gfc_2009_now: events 4; 180d avg 13.978%, median 14.201%, bullish 100.000%

1.8.1.3 jupiter_saturn_square

- **Events:** 27 total; 15 train, 12 test
- **Exact-date span:** 1935-11-26 → 2025-06-15
- **Mapped weekend/holiday events:** 8
- **Best horizon:** 60d
- **All events:** avg 4.658%, median 5.022%, bullish 77.778%; baseline avg 1.902%
- **Train edge:** +2.592 pp; **test edge:** +2.962 pp

Regime read:

1.8.1.4 jupiter_neptune_square

- **Events:** 26 total; 16 train, 10 test
- **Exact-date span:** 1929-06-07 → 2025-06-19
- **Mapped weekend/holiday events:** 12
- **Best horizon:** 90d
- **All events:** avg 5.235%, median 5.943%, bullish 84.615%; baseline avg 2.825%
- **Train edge:** +2.124 pp; **test edge:** +2.867 pp

Regime read:

- early_1927_1971: events 12; 90d avg 5.343%, median 5.003%, bullish 83.333%
- post_bretton_woods_1971_2008: events 9; 90d avg 3.091%, median 2.706%, bullish 77.778%
- post_gfc_2009_now: events 5; 90d avg 8.836%, median 7.813%, bullish 100.000%

1.8.1.5 pluto_saturn_sextile

- **Events:** 13 total; 6 train, 7 test
- **Exact-date span:** 1942-06-09 → 2026-03-28
- **Mapped weekend/holiday events:** 4
- **Best horizon:** 30d
- **All events:** avg 2.922%, median 1.332%, bullish 84.615%; baseline avg 0.951%
- **Train edge:** +0.885 pp; **test edge:** +2.902 pp

Regime read:

- early_1927_1971: events 4; 30d avg 3.290%, median 2.931%, bullish 100.000%
- post_bretton_woods_1971_2008: events 5; 30d avg 0.324%, median 0.003%, bullish 60.000%
- post_gfc_2009_now: events 4; 30d avg 5.800%, median 2.903%, bullish 100.000%

1.8.1.6 jupiter_pluto_square

- **Events:** 25 total; 15 train, 10 test
- **Exact-date span:** 1928-03-31 → 2023-05-17
- **Mapped weekend/holiday events:** 5
- **Best horizon:** 30d
- **All events:** avg 2.218%, median 1.390%, bullish 80.000%; baseline avg 0.951%
- **Train edge:** +0.582 pp; **test edge:** +2.293 pp

Regime read:

- early_1927_1971: events 11; 30d avg 0.493%, median 0.613%, bullish 63.636%
- post_bretton_woods_1971_2008: events 8; 30d avg 4.675%, median 4.869%, bullish 87.500%
- post_gfc_2009_now: events 6; 30d avg 2.103%, median 1.028%, bullish 100.000%

1.8.1.7 neptune_pluto_sextile

- **Events:** 45 total; 32 train, 13 test
- **Exact-date span:** 1945-02-19 → 2026-06-03
- **Mapped weekend/holiday events:** 14
- **Best horizon:** 90d
- **All events:** avg 4.620%, median 4.411%, bullish 75.000%; baseline avg 2.825%
- **Train edge:** +2.379 pp; **test edge:** +0.237 pp

Regime read:

- early_1927_1971: events 21; 90d avg 5.311%, median 5.250%, bullish 66.667%
- post_bretton_woods_1971_2008: events 17; 90d avg 3.702%, median 3.759%, bullish 82.353%
- post_gfc_2009_now: events 7; 90d avg 4.803%, median 7.224%, bullish 83.333%

1.8.2 Pressure candidates

1.8.2.1 neptune_saturn_trine

- **Events:** 16 total; 11 train, 5 test
- **Exact-date span:** 1929-03-06 → 2013-07-19
- **Mapped weekend/holiday events:** 4
- **Best horizon:** 180d
- **All events:** avg -3.238%, median -5.486%, bullish 43.750%; baseline avg 5.731%
- **Train edge:** -10.244 pp; **test edge:** -6.162 pp

Regime read:

- early_1927_1971: events 8; 180d avg -2.967%, median -3.054%, bullish 50.000%
- post_bretton_woods_1971_2008: events 5; 180d avg -12.204%, median -7.302%, bullish 0.000%
- post_gfc_2009_now: events 3; 180d avg 10.985%, median 11.175%, bullish 100.000%

1.8.2.2 jupiter_uranus_sextile

- **Events:** 22 total; 15 train, 7 test
- **Exact-date span:** 1929-08-07 → 2022-02-17
- **Mapped weekend/holiday events:** 6
- **Best horizon:** 90d
- **All events:** avg -3.206%, median -2.156%, bullish 36.364%; baseline avg 2.825%
- **Train edge:** -5.580 pp; **test edge:** -6.998 pp

Regime read:

- early_1927_1971: events 10; 90d avg -5.019%, median -3.717%, bullish 30.000%
- post_bretton_woods_1971_2008: events 9; 90d avg -1.762%, median -2.686%, bullish 33.333%
- post_gfc_2009_now: events 3; 90d avg -1.495%, median 3.470%, bullish 66.667%

1.8.2.3 saturn_uranus_square

- **Events:** 16 total; 10 train, 6 test
- **Exact-date span:** 1930-02-22 → 2022-10-04
- **Mapped weekend/holiday events:** 6
- **Best horizon:** 180d
- **All events:** avg -2.625%, median 2.254%, bullish 56.250%; baseline avg 5.731%
- **Train edge:** -11.837 pp; **test edge:** -2.552 pp

Regime read:

- early_1927_1971: events 7; 180d avg -9.499%, median 1.330%, bullish 57.143%
- post_bretton_woods_1971_2008: events 6; 180d avg 1.608%, median 0.244%, bullish 50.000%
- post_gfc_2009_now: events 3; 180d avg 4.951%, median 14.704%, bullish 66.667%

1.8.2.4 jupiter_pluto_opposition

- **Events:** 12 total; 7 train, 5 test
- **Exact-date span:** 1937-05-29 → 2014-01-31
- **Mapped weekend/holiday events:** 6
- **Best horizon:** 90d
- **All events:** avg -2.851%, median -1.227%, bullish 50.000%; baseline avg 2.825%
- **Train edge:** -5.171 pp; **test edge:** -6.384 pp

Regime read:

- early_1927_1971: events 6; 90d avg -2.284%, median -2.414%, bullish 50.000%
- post_bretton_woods_1971_2008: events 4; 90d avg -8.636%, median -7.611%, bullish 25.000%
- post_gfc_2009_now: events 2; 90d avg 7.020%, median 7.020%, bullish 100.000%

1.8.2.5 jupiter_uranus_square

- **Events:** 24 total; 16 train, 8 test
- **Exact-date span:** 1930-09-05 → 2021-01-17
- **Mapped weekend/holiday events:** 7
- **Best horizon:** 180d
- **All events:** avg -1.260%, median 1.386%, bullish 54.167%; baseline avg 5.731%
- **Train edge:** -8.443 pp; **test edge:** -4.085 pp

Regime read:

- early_1927_1971: events 11; 180d avg -5.242%, median 2.620%, bullish 54.545%
- post_bretton_woods_1971_2008: events 10; 180d avg -0.829%, median -2.388%, bullish 40.000%
- post_gfc_2009_now: events 3; 180d avg 11.908%, median 14.346%, bullish 100.000%

1.8.2.6 pluto_saturn_trine

- **Events:** 13 total; 7 train, 6 test
- **Exact-date span:** 1937-03-26 → 2008-04-26
- **Mapped weekend/holiday events:** 5
- **Best horizon:** 180d
- **All events:** avg 0.392%, median 3.794%, bullish 61.538%; baseline avg 5.731%
- **Train edge:** -8.987 pp; **test edge:** -1.081 pp

Regime read:

- early_1927_1971: events 6; 180d avg -5.082%, median 0.123%, bullish 50.000%
- post_bretton_woods_1971_2008: events 7; 180d avg 5.084%, median 7.695%, bullish 71.429%
- post_gfc_2009_now: events 0; 180d avg n/a, median n/a, bullish n/a

1.8.2.7 jupiter_uranus_trine

- **Events:** 27 total; 16 train, 11 test
- **Exact-date span:** 1931-10-10 → 2019-12-15
- **Mapped weekend/holiday events:** 11
- **Best horizon:** 180d
- **All events:** avg 0.712%, median 3.020%, bullish 62.963%; baseline avg 5.731%
- **Train edge:** -7.017 pp; **test edge:** -2.112 pp

Regime read:

- early_1927_1971: events 12; 180d avg -0.426%, median 5.613%, bullish 58.333%
- post_bretton_woods_1971_2008: events 11; 180d avg 1.362%, median 0.852%, bullish 72.727%
- post_gfc_2009_now: events 4; 180d avg 2.336%, median 1.717%, bullish 50.000%

1.9 Theme-level read

Feature individual bisa kuat tapi sample kecil. Theme bucket menggabungkan beberapa feature terkait untuk melihat apakah narasi lebih besar tahan. Ini lebih cocok untuk teori kerja daripada entry/exit signal.

1.9.1 Outer Big Cycles — anchored themes

#	Theme / feature bucket	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Included features
1	jupiter_uranus_major	100	180d	1.073%	3.541%	60.000%	-6.162 pp	-2.095 pp	jupiter_uranus_conjunct, jupiter_uranus_opposite, jupiter_uranus_sextile, jupiter_uranus_square, jupiter_uranus_trine
2	saturn_pluto_major	53	60d	-0.922%	0.960%	55.769%	-4.974 pp	-0.111 pp	pluto_saturn_conjunct, pluto_saturn_opposite, pluto_saturn_sextile, pluto_saturn_square, pluto_saturn_trine
3	jupiter_saturn_major	101	60d	2.769%	3.310%	74.257%	+0.467 pp	+1.430 pp	jupiter_saturn_conjunct, jupiter_saturn_opposite, jupiter_saturn_sextile, jupiter_saturn_square, jupiter_saturn_trine

1.9.2 No-Moon Full Scan — anchored themes

#	Theme / feature bucket	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Included features
1	saturn_pluto_major	52	180d	0.709%	2.700%	58.824%	-8.531 pp	-0.008 pp	pluto_saturn_conjuncti pluto_saturn_oppositi pluto_saturn_sextile, pluto_saturn_square, pluto_saturn_trine
2	jupiter_uranus_major	100	180d	1.492%	4.121%	61.000%	-5.589 pp	-1.941 pp	jupiter_uranus_conjunc jupiter_uranus_oppos. jupiter_uranus_sexti jupiter_uranus_squar jupiter_uranus_trine
3	jupiter_saturn_major	101	60d	2.781%	3.310%	72.277%	+0.509 pp	+1.422 pp	jupiter_saturn_conjunc jupiter_saturn_oppos. jupiter_saturn_sexti jupiter_saturn_squar jupiter_saturn_trine
4	saturn_personal_conjuncti on_pressure	167	30d	0.329%	0.993%	56.131%	-0.795 pp	-0.340 pp	mars_saturn_conjuncti mercury_saturn_conjunc saturn_sun_conjuncti saturn_venus_conjuncti
5	mercury_venus_cooldown	461	14d	0.174%	0.578%	56.182%	-0.262 pp	-0.272 pp	mercury_venus_conjunc mercury_venus_sextile
6	jupiter_inner_boost	1030	30d	0.595%	1.111%	59.515%	-0.542 pp	-0.065 pp	jupiter_mars_conjuncti jupiter_mars_trine, jupiter_mercury_conj jupiter_mercury_trine jupiter_sun_conjuncti
7	mars_inner_trigger_all	1057	3d	0.270%	0.284%	57.994%	+0.089 pp	+0.310 pp	+3 mars_mercury_conjuncti mars_mercury_sextile, mars_mercury_trine, mars_sun_conjunction, mars_sun_sextile
8	venus_opposition_riskon	441	14d	0.217%	0.689%	57.370%	-0.237 pp	-0.203 pp	+4 jupiter_venus_oppositi mars_venus_opposition pluto_venus_oppositi saturn_venus_oppositi uranus_venus_oppositi
9	uranus_hard_to_personal	1449	60d	1.659%	2.372%	62.932%	-0.354 pp	-0.068 pp	mars_uranus_conjuncti mars_uranus_oppositi mars_uranus_square, mercury_uranus_conjunc mercury_uranus_oppos. +7

#	Theme / feature bucket	Events	Best horizon	Avg	Median	Bullish	Train edge	Test edge	Included features
10	saturn_hard_to_personal	1418	3d	0.185%	0.216%	57.052%	+0.032 pp	+0.183 pp	mars_saturn_conjuncti mars_saturn_oppositi mars_saturn_square, mercury_saturn_conjun mercury_saturn_oppos. +7
11	pluto_hard_to_personal	1474	7d	0.093%	0.302%	54.817%	-0.197 pp	-0.013 pp	mars_pluto_conjuncti mars_pluto_oppositio mars_pluto_square, mercury_pluto_conjun mercury_pluto_opposi +7
12	mars_inner_trigger_soft	610	3d	0.223%	0.272%	58.689%	-0.009 pp	+0.341 pp	mars_mercury_conjunc mars_mercury_trine, mars_sun_conjunction, mars_sun_trine, mars_venus_conjuncti +1

1.10 Interpretasi radikal tapi disiplin

1.10.1 Saturn-Neptune: square konstruktif, trine pressure

Hasil paling provokatif adalah split Saturn-Neptune. Square tampil sebagai repair/re-rating window yang kuat di 180d, sementara trine tampil sebagai underperformance candidate. Ini tidak cocok dengan pembacaan simbolik dangkal trine baik, square buruk. Secara data, konfigurasi tension justru bisa bertepatan dengan fase penyesuaian yang kemudian direprice positif, sedangkan harmoni Saturn-Neptune bisa bertepatan dengan stagnasi atau disillusionment market.

1.10.2 Jupiter tidak selalu risk-on

Jupiter-Uranus conjunction terlihat konstruktif, tapi Jupiter-Uranus sextile justru pressure candidate. Jupiter-Saturn sextile konstruktif, sementara Jupiter-Saturn opposition lemah. Jadi yang terbaca bukan planet baik/buruk, melainkan kombinasi pair + aspect + horizon.

1.10.3 SPX berbeda dari BTC

BTC cenderung lebih sensitif terhadap release/risk-on yang cepat. SPX, karena komposisi large-cap, earnings cycle, buyback, dan policy regime, punya drift yang membuat banyak event positif nominal tapi tidak outperformance. Maka theme yang tampak bullish di BTC bisa menjadi underperformance relatif di SPX.

1.10.4 Personal planets sebagai trigger, bukan siklus utama

Mars/Sun/Mercury/Venus feature yang konstruktif pada 180d kemungkinan bukan penyebab, melainkan marker fase market yang lebih besar. Ini butuh validasi tambahan karena no-moon full scan punya multiple testing jauh lebih berat.

1.11 Anti-overfitting notes

- Multiple testing sangat besar: 24 feature di outer scan dan 145 feature di no-moon full scan, plus horizon dan theme bucket.
- Sample aspek planet luar kecil; angka terlihat ekstrem bisa berubah setelah beberapa dekade data baru.
- Average bisa didominasi crash/rebound; median dan bullish rate wajib dilihat bersama.
- Train/test split kronologis membantu, tapi bukan pengganti pre-registration.
- Mapping weekend/libur ke trading day berikutnya membuat penelitian dapat dieksekusi di pasar saham, tapi dapat menggeser interpretasi astrologis 1–3 hari.
- ^GSPC bukan tradable instrument; perlu validasi di SPY, ES futures continuous, Dow, Nasdaq, equal-weight S&P, dan international equities.

1.12 Working theory untuk SPX

1. **Constructive expansion / repair:** Saturn–Neptune square, Jupiter–Saturn sextile, Jupiter–Uranus conjunction.
2. **Pressure / underperformance:** Saturn–Neptune trine, Jupiter–Uranus sextile, Saturn–Uranus square, Jupiter–Saturn opposition.
3. **Cooldown:** Mercury–Venus conjunction/sextile bucket menunjukkan mild underperformance di 14d, konsisten train/test.
4. **BTC-to-SPX transfer:** Venus opposition risk-on tidak transfer bersih; SPX justru short-horizon underperformance relatif baseline.
5. **Use-case praktis:** bukan buy/sell, tapi watchlist kalender untuk menguji apakah window tertentu bertepatan dengan re-rating, compression, atau volatility clustering.

1.13 Next research steps

- Re-run exact same hypothesis on SPY, ES=F / continuous futures if available, ^DJI, ^IXIC, RSP, and global equity indices.
- Freeze top rules from this paper, then test forward on 2026–2028 without re-optimizing.
- Add bootstrap/permutation test: compare exact aspect dates against random dates matched by decade/month/weekday.
- Add drawdown and volatility metrics, not only forward return.
- Build future watch calendar per theory category, clearly labeled as hypothesis watchlist.

1.14 Appendix A. Output artifacts

- Raw JSON: /tmp/radical-spx500-astro-search-v011.json
- Markdown paper: /tmp/spx500-radical-astrology-research-hermetic-alpha-v011.md
- PDF paper: /tmp/spx500-radical-astrology-research-hermetic-alpha-v011.pdf

1.15 Disclaimer

This is exploratory research and library dogfooding only. It is not investment advice, not a trading system, and not evidence of astrological causality. Any market use requires independent validation, risk management, and pre-registered forward testing.